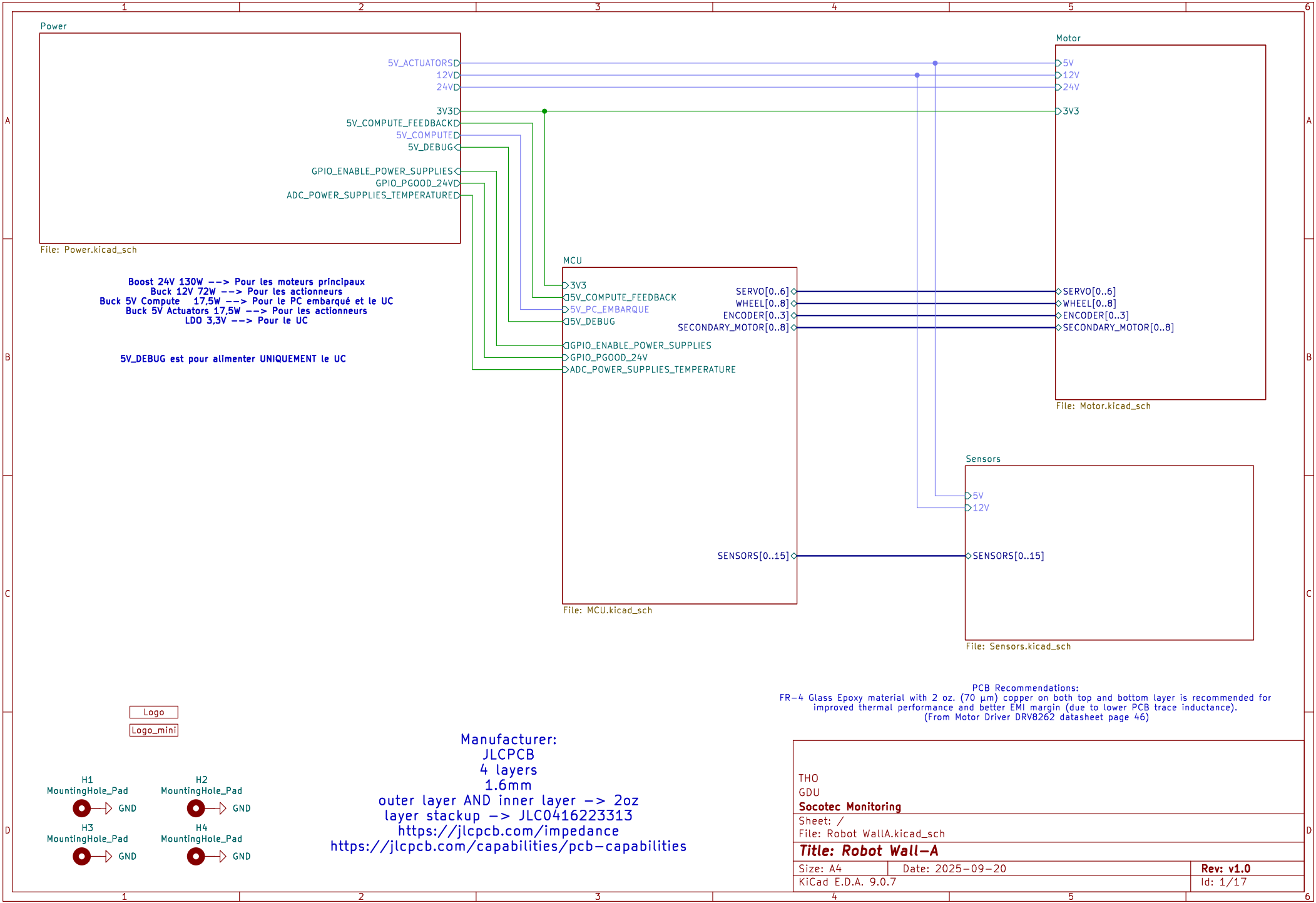




Boost 24V 130W --> Pour les moteurs principaux
Buck 12V 72W --> Pour les actionneurs
Buck 5V Compute 17,5W --> Pour le PC embarqué et le UC
Buck 5V Actuators 17,5W --> Pour les actionneurs
LDO 3,3V --> Pour le UC

5V_DEBUG est pour alimenter UNIQUEMENT le UC

Logo
Logo_mini



Manufacturer:
JLCPCB
4 layers
1.6mm
outer layer AND inner layer --> 2oz
layer stackup --> JLC0416223313
<https://jlcpcb.com/impedance>
<https://jlcpcb.com/capabilities/pcb-capabilities>

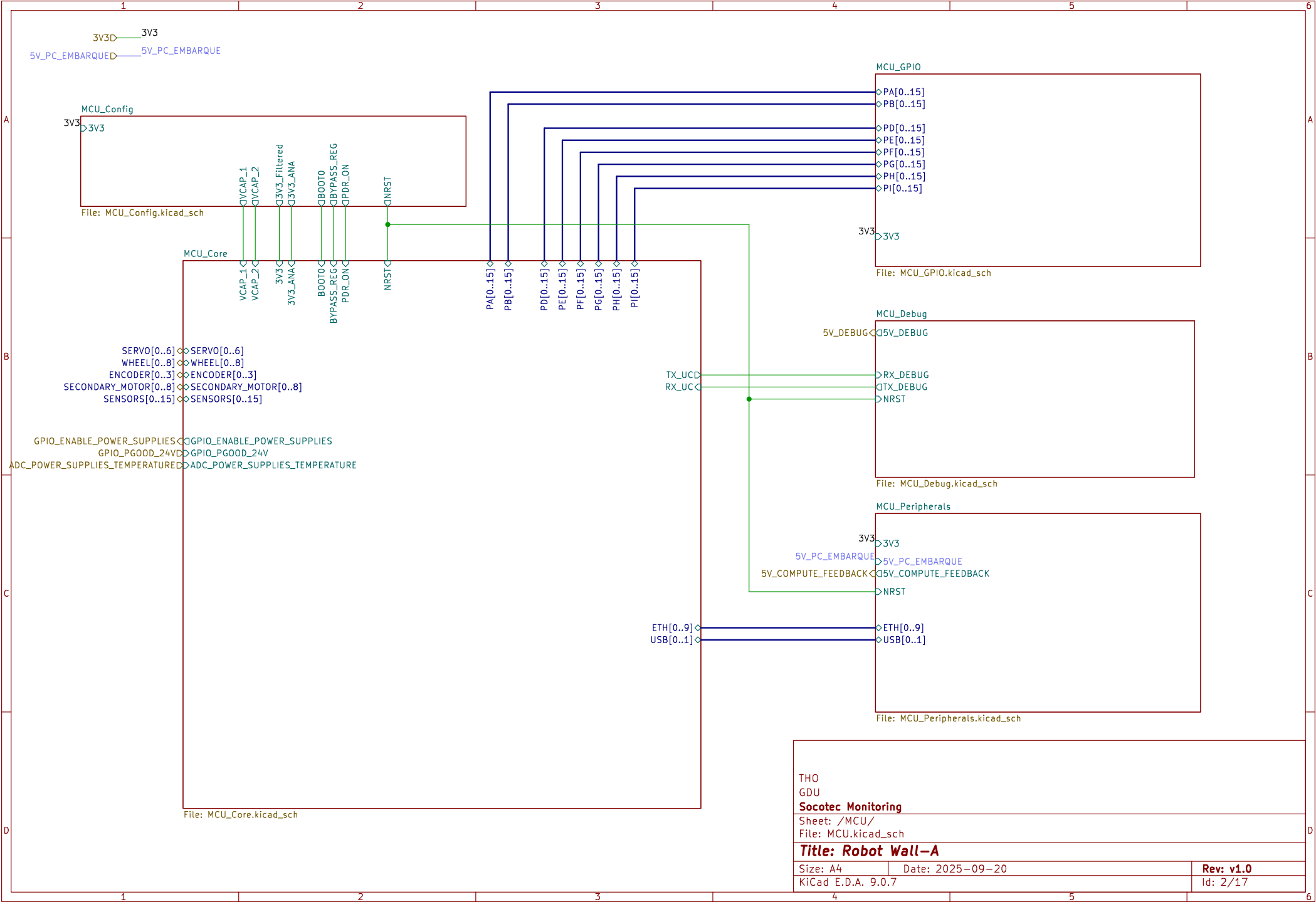
PCB Recommendations:
FR-4 Glass Epoxy material with 2 oz. (70 μm) copper on both top and bottom layer is recommended for improved thermal performance and better EMI margin (due to lower PCB trace inductance).
(From Motor Driver DRV8262 datasheet page 46)

TH0
GDU
Socotec Monitoring
Sheet: /
File: Robot WallA.kicad_sch

Title: Robot Wall-A

Size: A4
Date: 2025-09-20
KiCad E.D.A. 9.0.7

Rev: v1.0
Id: 1/17



TH0

GDU

Socotec Monitoring

Sheet: /MCU/

File: MCU.kicad_sch

Title: Robot Wall-A

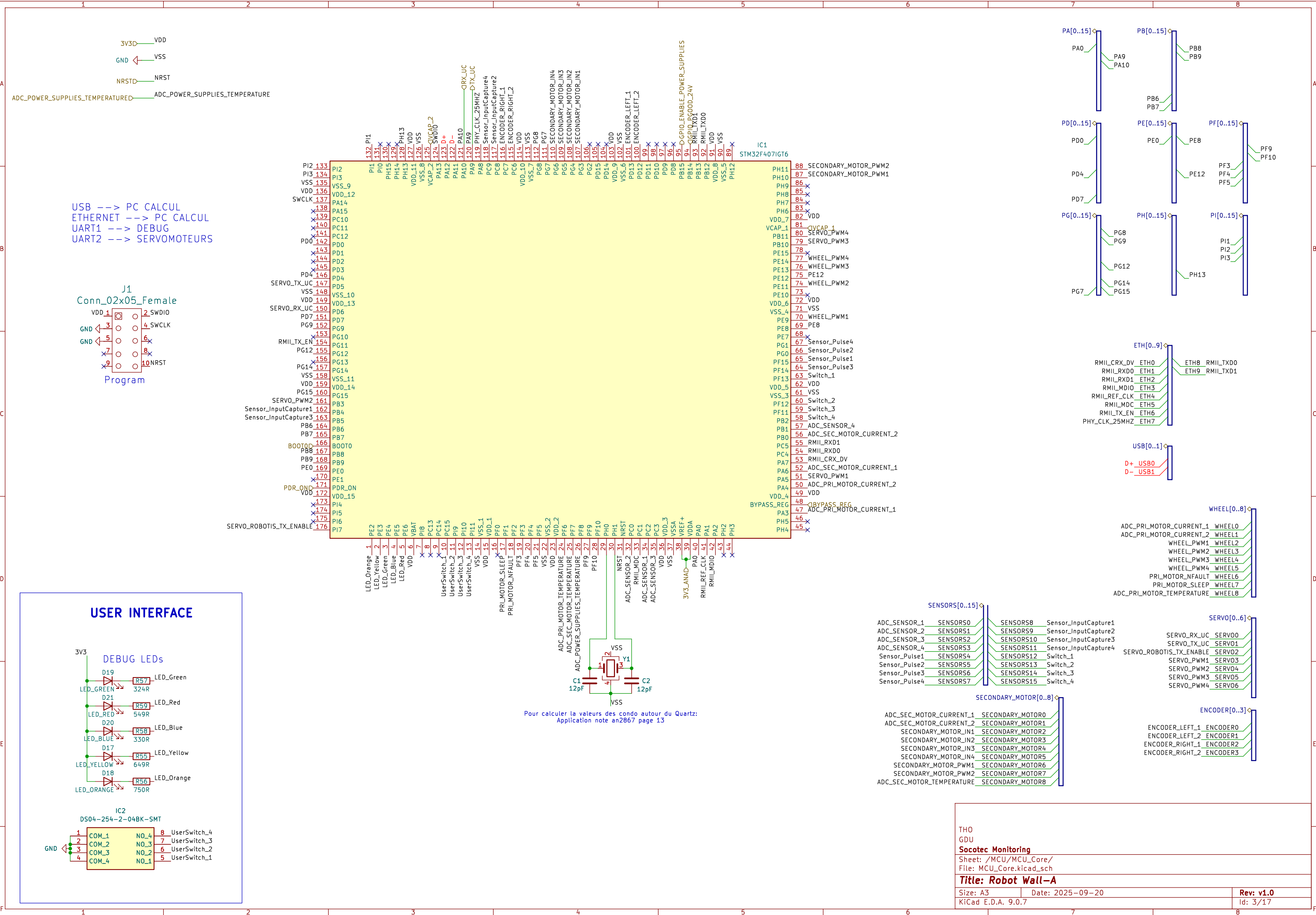
Size: A4

Date: 2025-09-20

Rev: v1.0

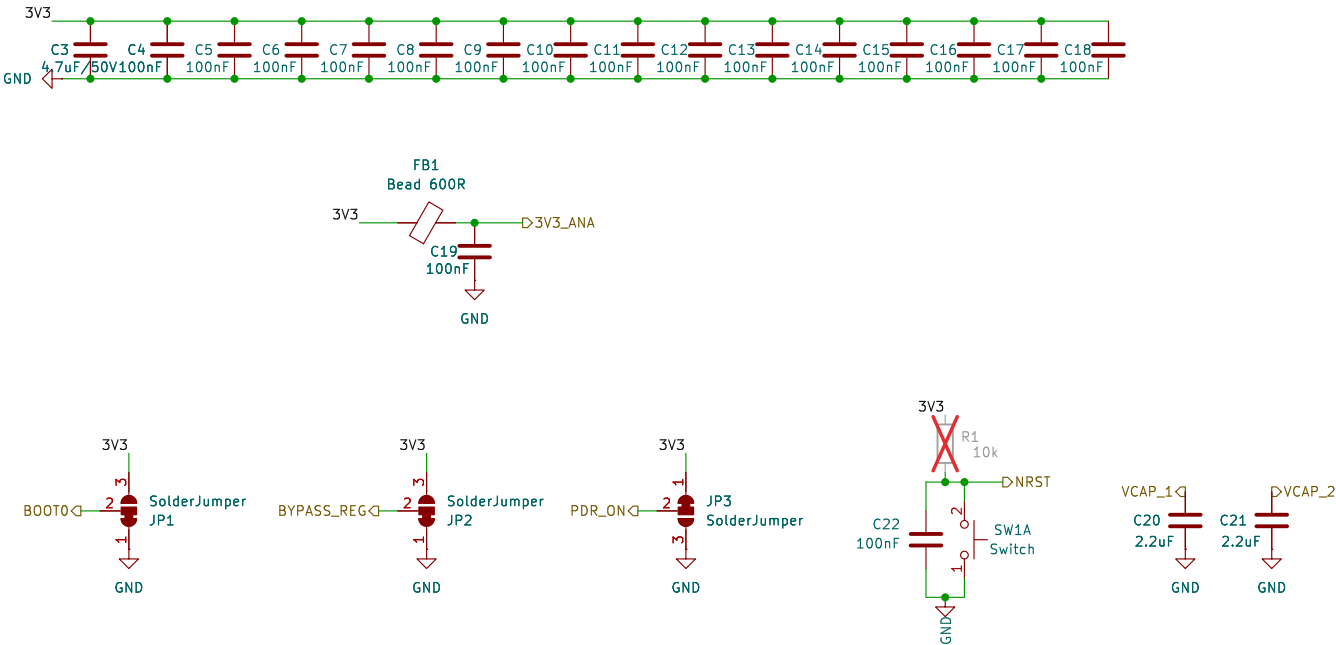
KiCad E.D.A. 9.0.7

Id: 2/17



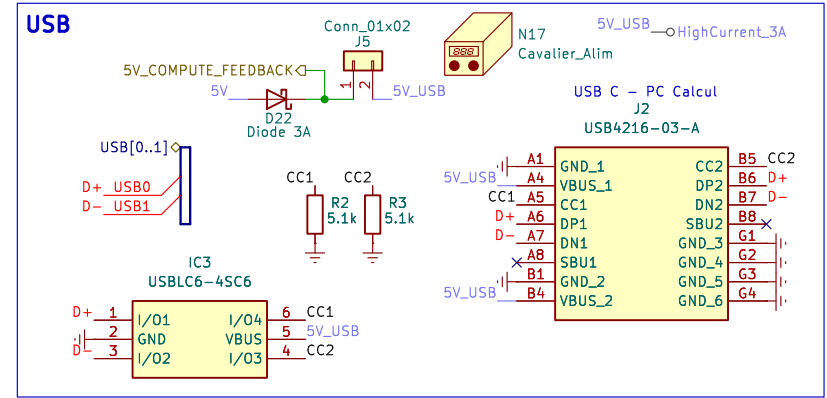
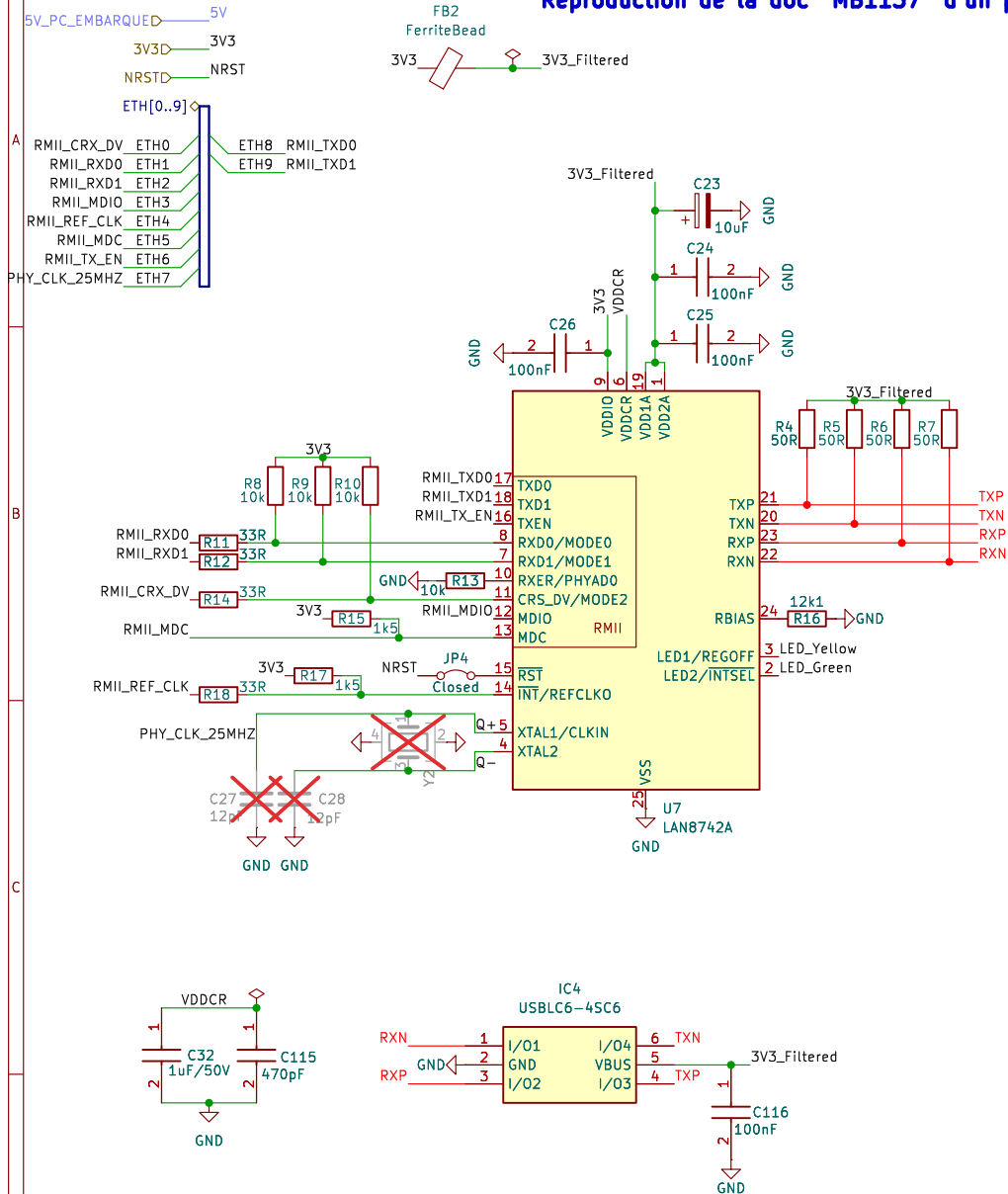
Must be as close as possible from μC

3V3 $\rightarrow$ 3V3_Filtered

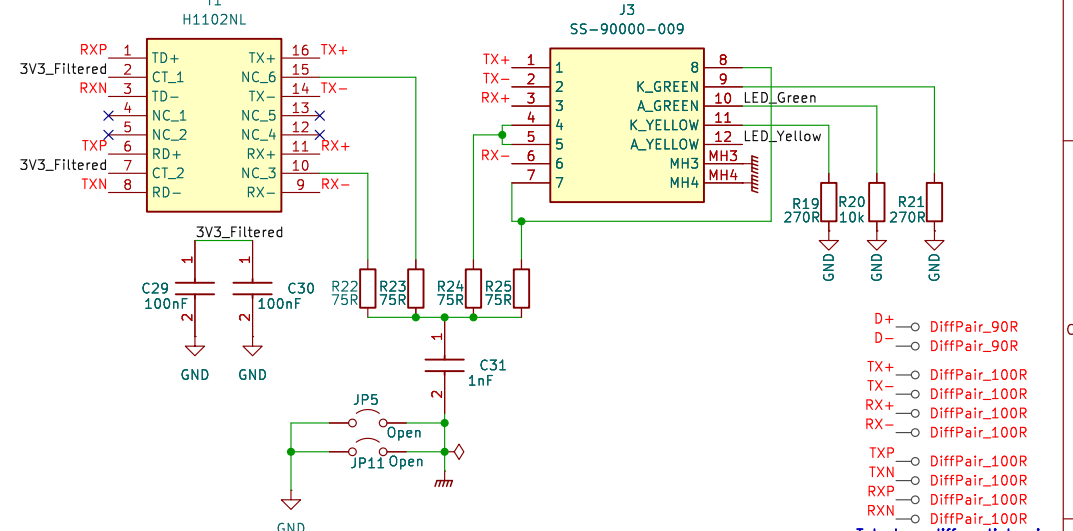


TH0		
GDU		
Socotec Monitoring		
Sheet: /MCU/MCU_Config/		
File: MCU_Config.kicad_sch		
Title: Robot Wall-A		
Size: A4	Date: 2025-09-20	Rev: v1.0
KiCad E.D.A. 9.0.7	Id: 4/17	

Reproduction de la doc "MB1137" d'un µC STM32 (pdf dans le git, dossier Docs)



TX peut etre connecté sur RX car le module est compatible auto-MDIX



Tuto tune differential pairs
<https://www.youtube.com/watch?v=M13QxtPvXYt&t=798s>

TH0
 GDU

Socotec Monitoring

Sheet: /MCU/MCU_Peripherals/
 File: MCU_Peripherals.kicad_sch

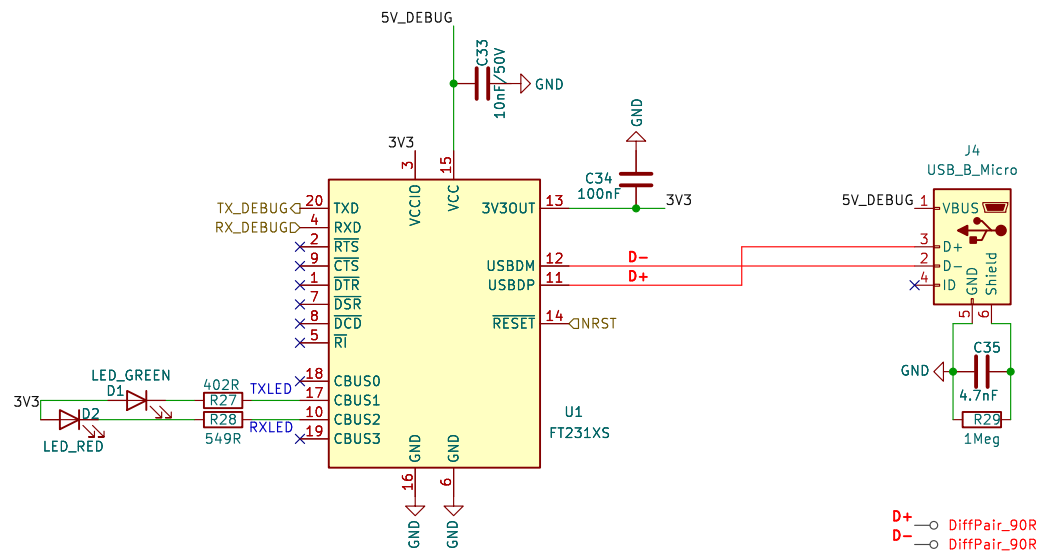
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.7

Rev: v1.0

Id: 5/17



TH0
GDU

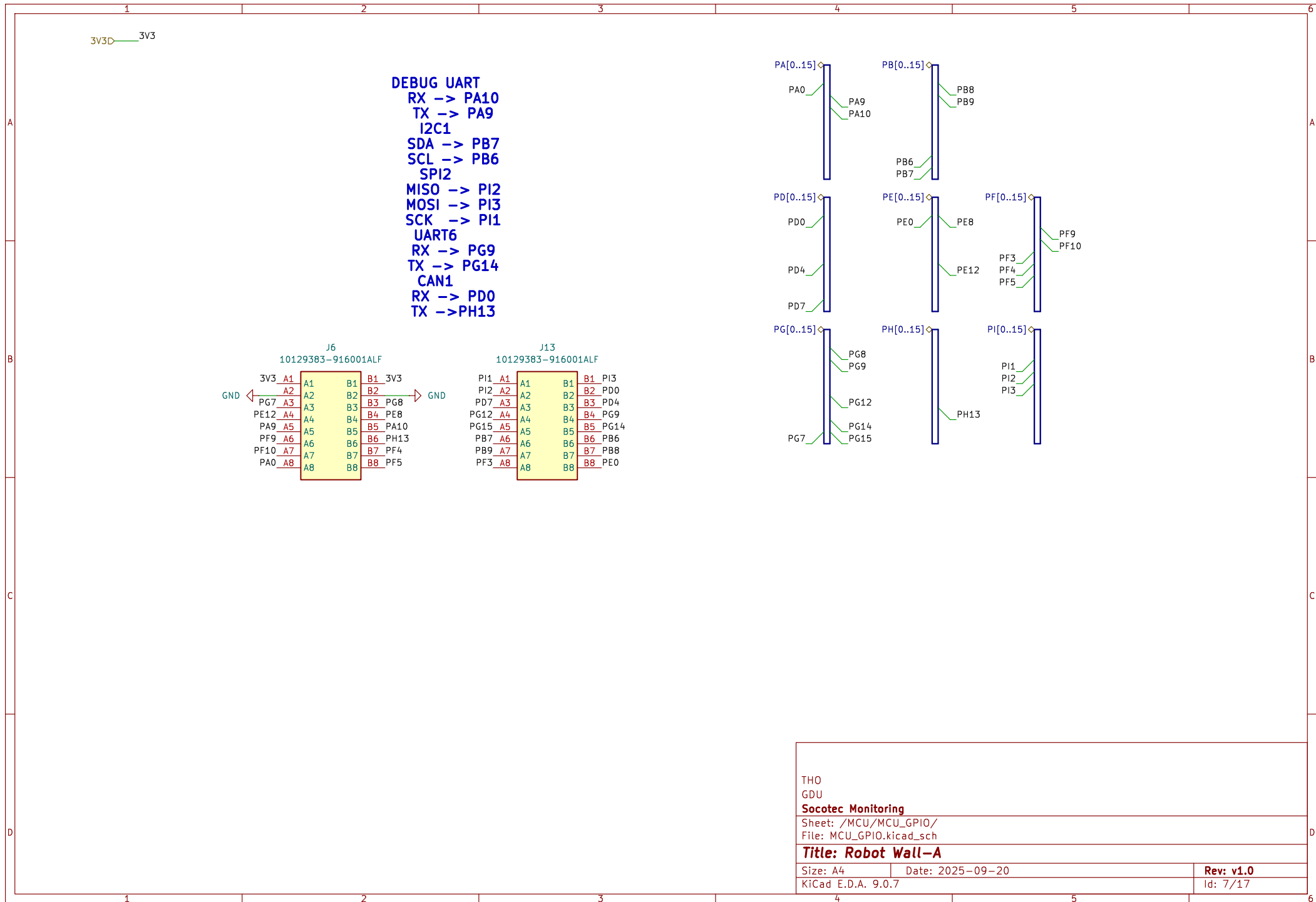
Socotec Monitoring

Sheet: /MCU/MCU_Debug/
File: MCU_Debug.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20
KiCad E.D.A. 9.0.7

Rev: v1.0
Id: 6/17



TH0
GDU

Socotec Monitoring

Sheet: /MCU/MCU_GPIO/
File: MCU_GPIO.kicad_sch

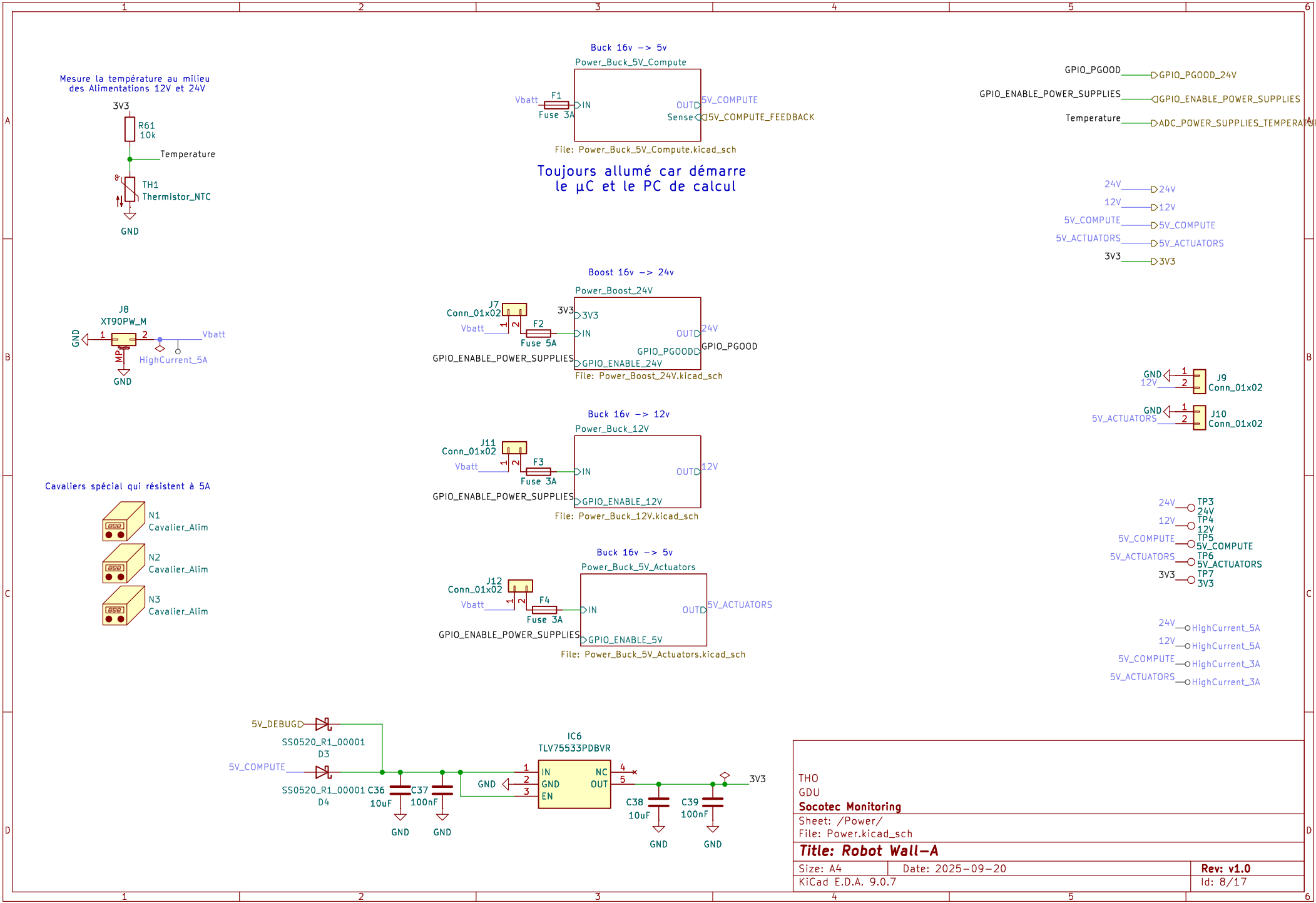
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.7

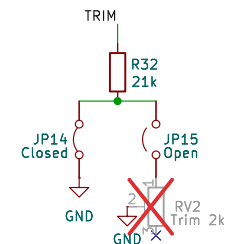
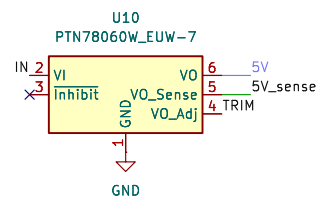
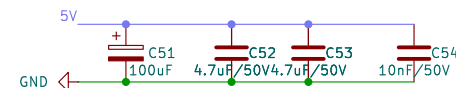
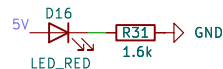
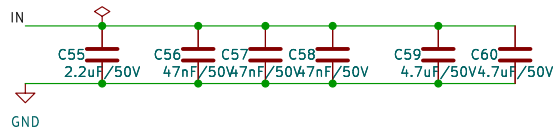
Rev: v1.0

Id: 7/17



IND IN

5V OUT
5V_sense Sense



TH0
GDU

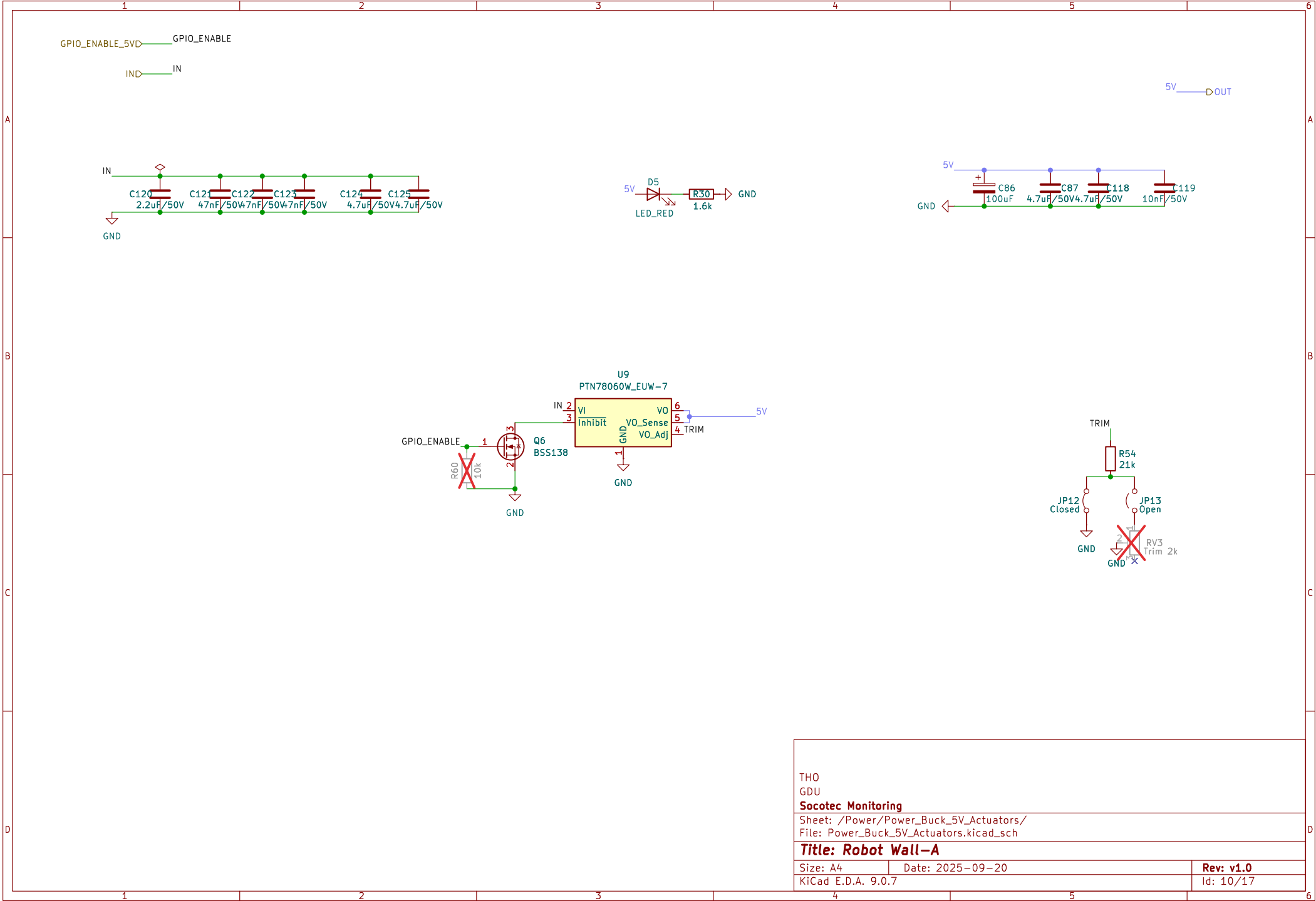
Socotec Monitoring

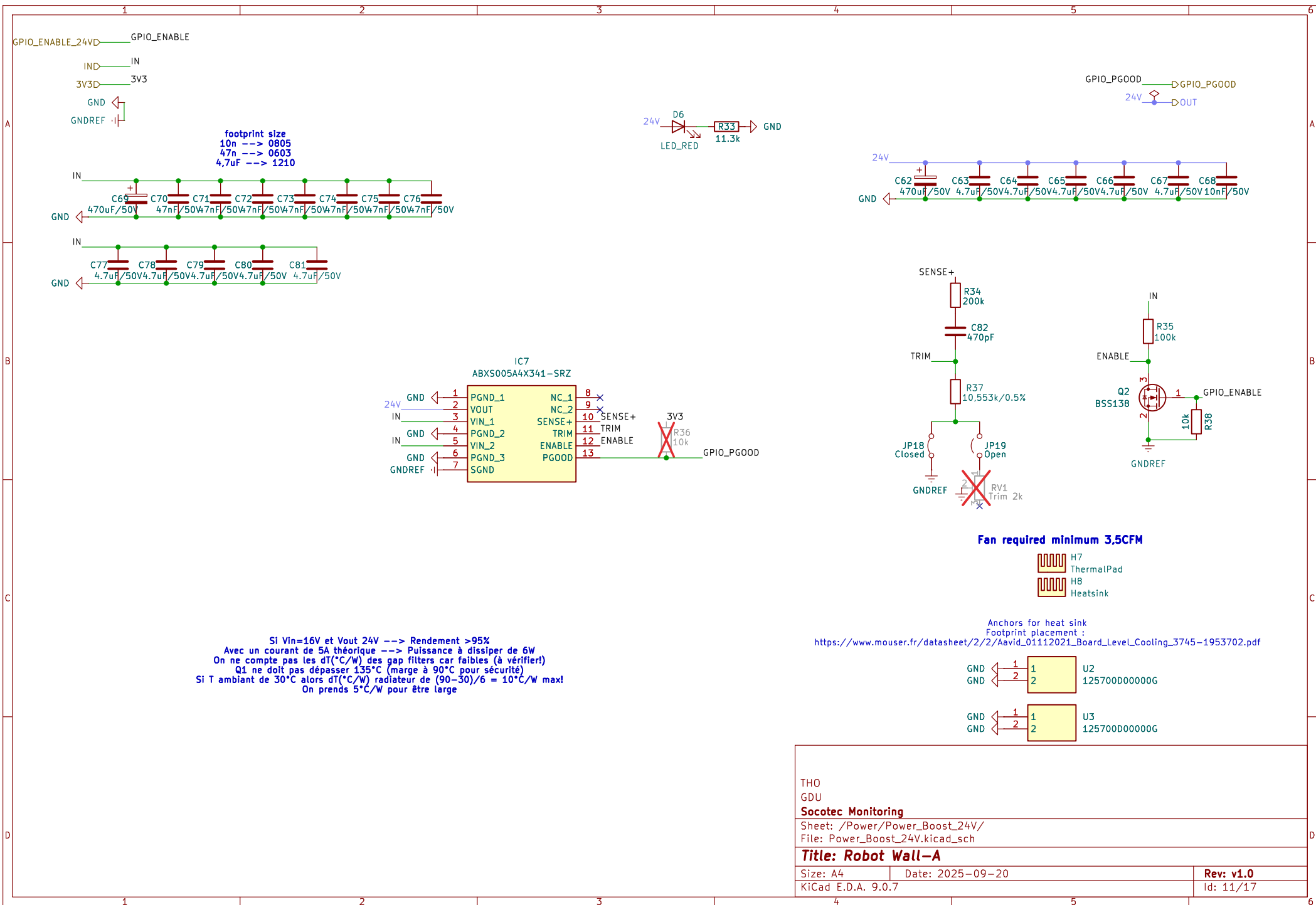
Sheet: /Power/Power_Buck_5V_Compute/
File: Power_Buck_5V_Compute.kicad_sch

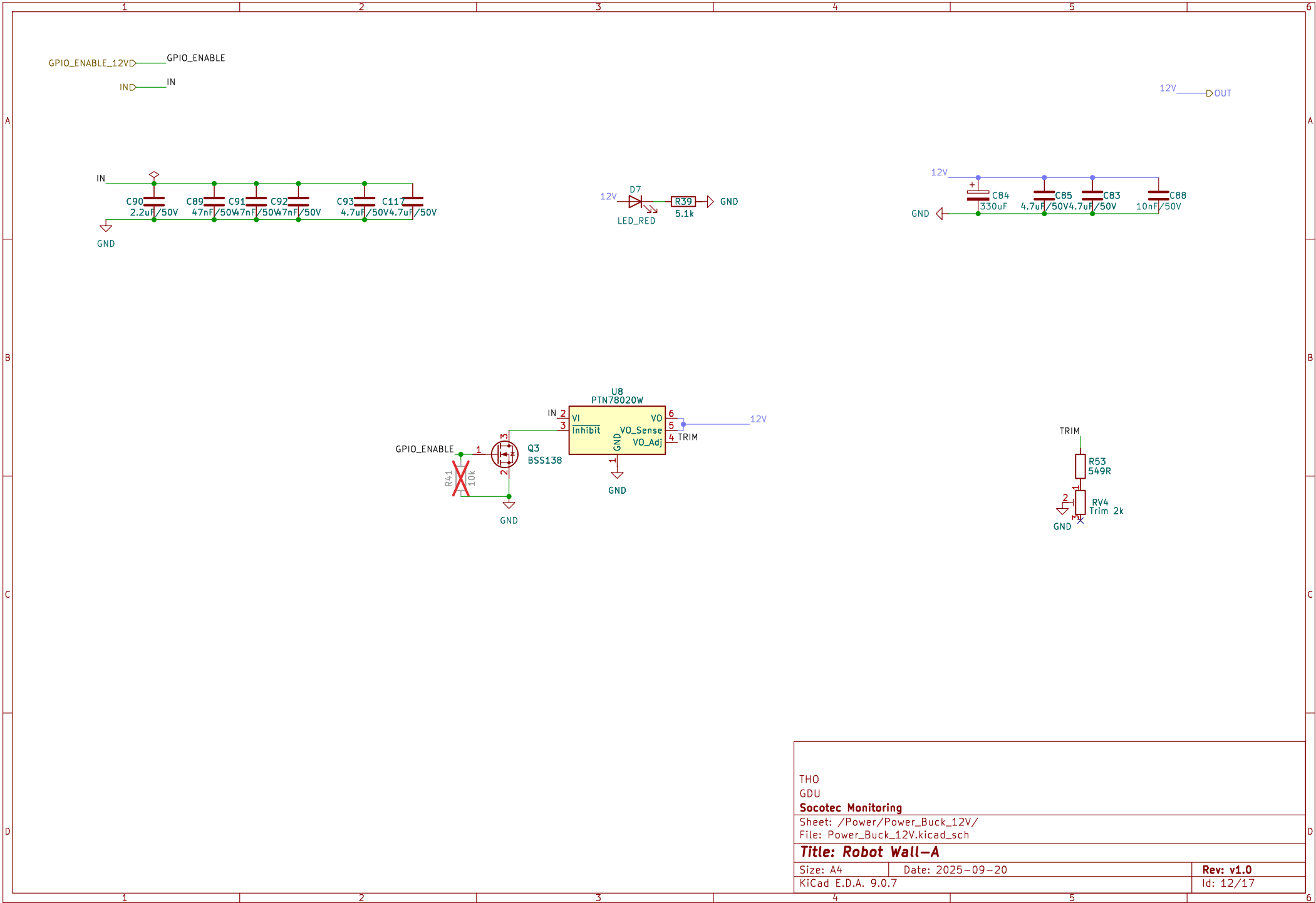
Title: Robot Wall-A

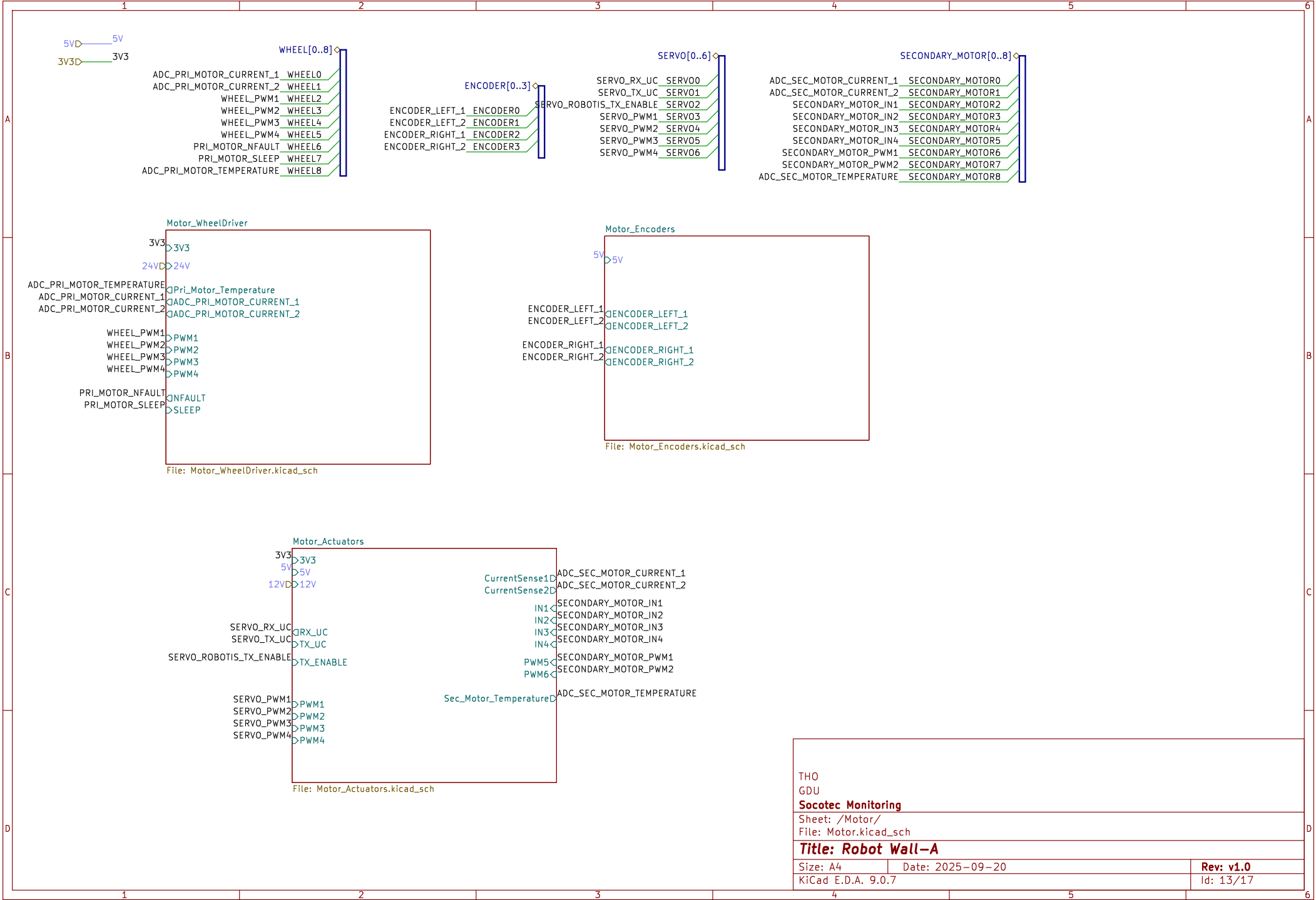
Size: A4 Date: 2025-09-20
KiCad E.D.A. 9.0.7

Rev: v1.0
Id: 9/17







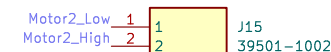
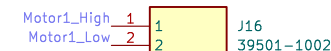
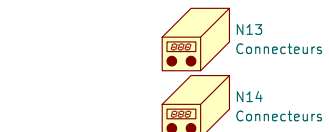
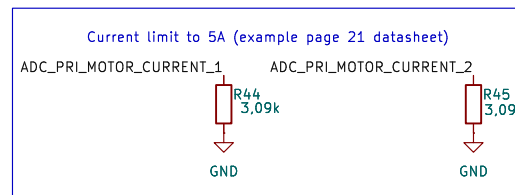
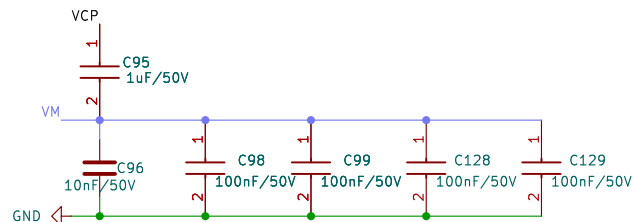
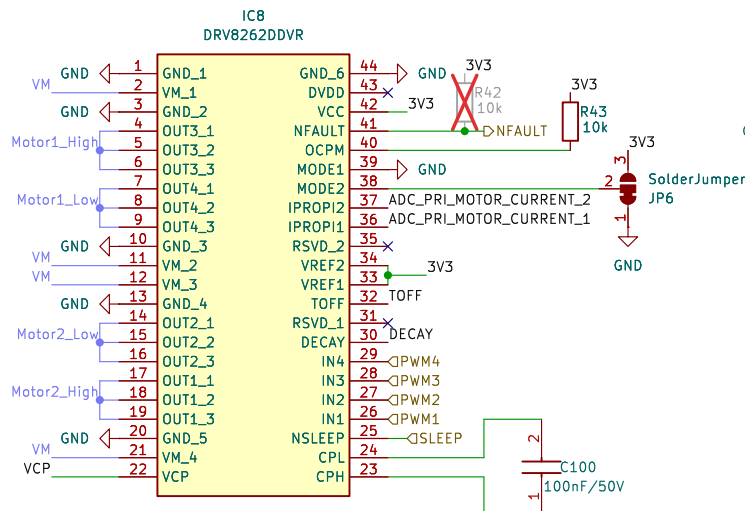


3V3D — 3V3
24VD — VM



Default Configuration:
OCPM = 1 --> Over Current Protection error = Retry
MODE = 0 --> PH/EN mode
Toff = 1 --> Waiting time between change = 16µs
Decay = HZ --> Auto detect mode

ADC_PRI_MOTOR_CURRENT_1 —> ADC_PRI_MOTOR_CURRENT_1
ADC_PRI_MOTOR_CURRENT_2 —> ADC_PRI_MOTOR_CURRENT_2



ATTENTION!!
Pour simplifier le routage,
les phases du moteurs on été inversées.
Attention au câblage!

https://www.mouser.fr/datasheet/2/2/Aavid_01112021_Board_LeveL_Cooling_3745-1953702.pdf

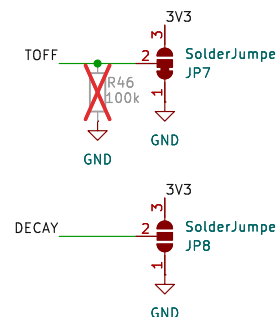


Fan required minimum 3,5CFM

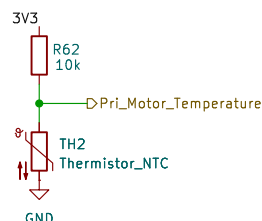
2x Transmotec PD4266-24-17-BFEC
24V / 2,1A nominal
4,2A nominal total

Power loss calculation and dissipator estimation
(page 33 to 35 datasheet)
Calculations already done in datasheet (same example with 4A)
Conclusion: Dissipator with less than 4°C/W

Heat sink possibilities:
WAVE-35-12 with anchors 125700D00000G --> Fan required minimum 3CFM
374524B60023G with anchors 125700D00000G --> Fan required minimum 3,5CFM
567-45AB with screws --> Fan required minimum 4CFM



Motor1_High —> HighCurrent_5A
Motor1_Low —> HighCurrent_5A
Motor2_High —> HighCurrent_5A
Motor2_Low —> HighCurrent_5A



TH0
GDU

Socotec Monitoring

Sheet: /Motor/Motor_WheelDriver/
File: Motor_WheelDriver.kicad_sch

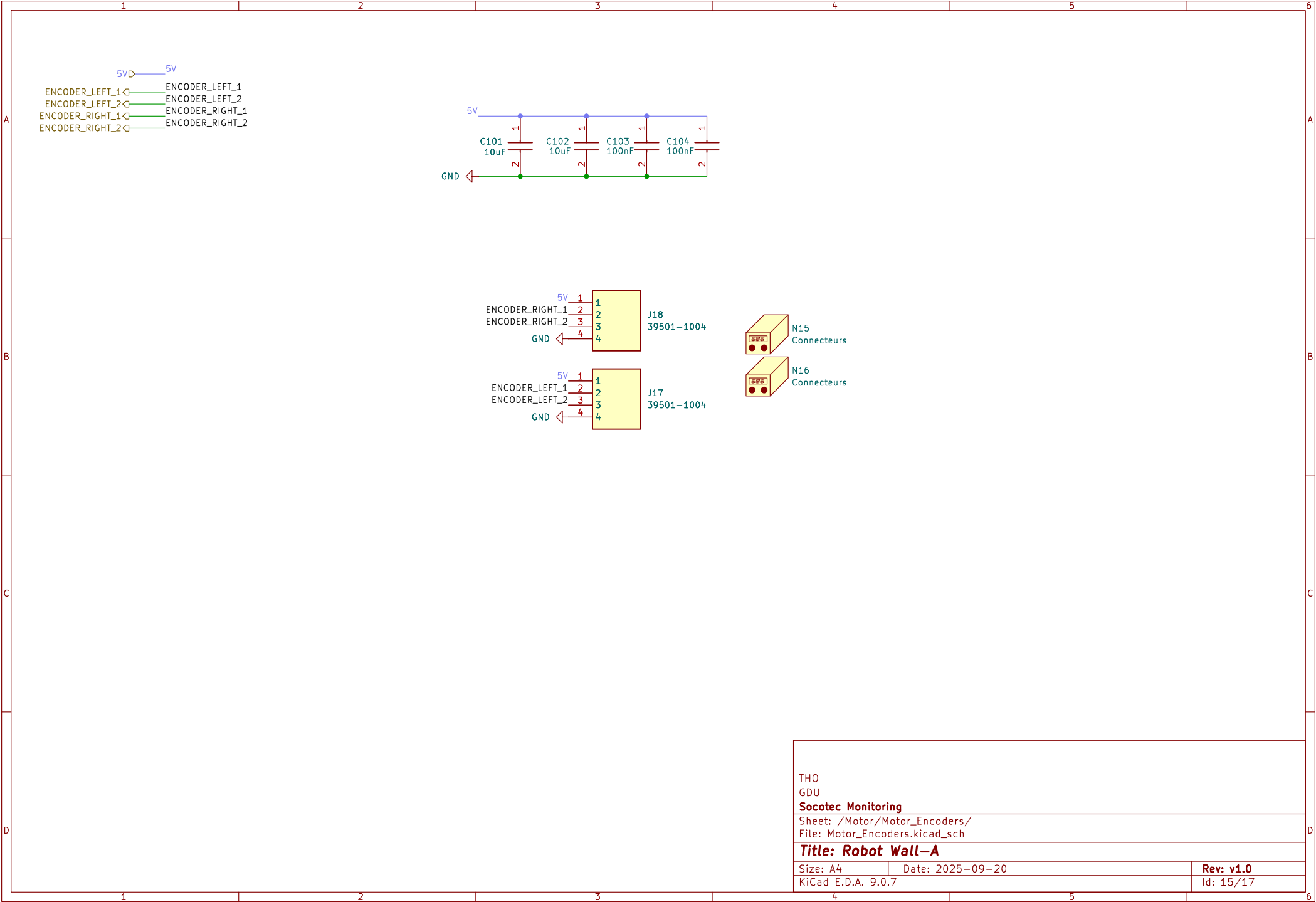
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.7

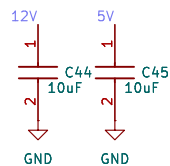
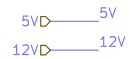
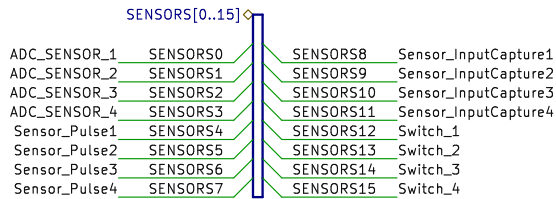
Rev: v1.0

Id: 14/17



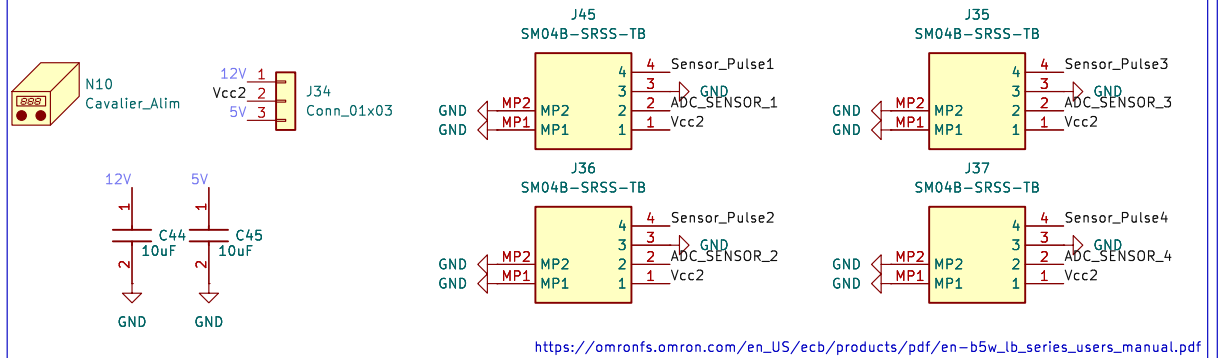


Id: 16/17

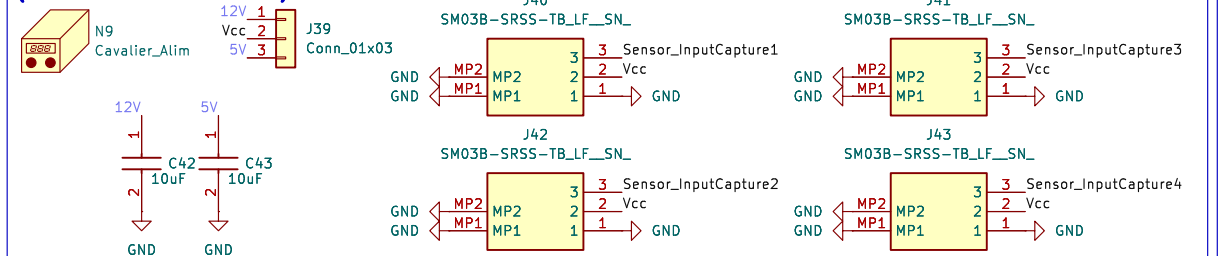


Capteurs de distance

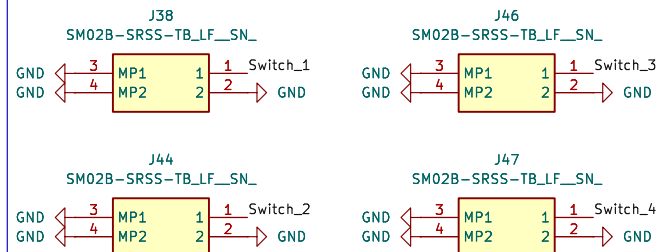
Pour capteur B5W-LB2101-1 (cable A04ZR04ZR28H305A 30cm)
analogique 5V



Pour capteur Pololu
<https://www.pololu.com/product/5464>
<https://www.pololu.com/product/5472>
(cable reference: 5517)



Interrupteurs standard



TH0
GDU

Socotec Monitoring

Sheet: /Sensors/
File: Sensors.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.7

Rev: v1.0

Id: 17/17